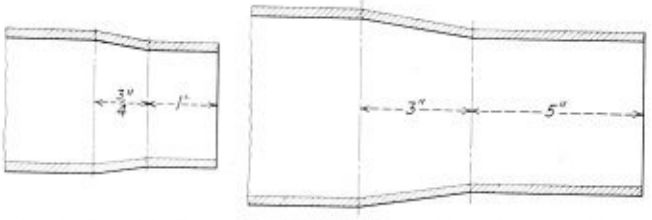
In the preparation of firebox tube sheets, tube holes in the firebox tube sheet should be finished 1/64 inch larger than the outside diameter of the ferrule to be used. All holes in firebox tube sheets for tubes or arch pipes should be reamed with a straight reamer to exact size and the sharp edges removed. The edges of the hole should be finished to a radius of approximately $\frac{1}{16}$ inch to prevent the cutting of the tubes. It is highly important that no sharp edges be left on the sheet, and both workman and inspector should see that the holes are properly prepared in this respect.

In the preparation of smokebox tube sheets, tube holes should be finished 1/32 inch larger in diameter than the outside diameter of the tube end and the tube holes must be cleaned and sharp edges removed in the same manner as recommended for firebox tube sheets. The contour of the edges is equally important for the smokebox tube sheet.

In the preparation of tubes, the firebox end of the body tube in ordinary Baldwin practice is swaged down toward its end, this swaging dimension being usually




**Figs. 2 and 3.—Swaged ends of body and superheater tubes**

$\frac{1}{8}$ inch below the nominal diameter of the tube. To permit the end of the tube to fit the ferrule neatly after the ferrule has been first tightened in the hole in the tube sheet, the swaged end of the tube exclusive of the tapered portion should not be less than 1 inch long and in general form as shown by Fig. 2. This length of swage allows $\frac{1}{4}$ inch for beading beyond the edge of a tube sheet $\frac{1}{2}$ inch in thickness. Any increase in the thickness of the tube sheet will necessitate a proportionate increase in the width of the ferrule and the length of the swaging. After swaging, the ends of the tubes should have all scale removed by light filing or grinding.

The firebox end of the superheater tube in ordinary Baldwin practice is swaged down to an average outside diameter of $4\frac{1}{2}$ inches, if the nominal diameter of the tube is $5\frac{1}{4}$ inches, $5\frac{3}{8}$ inches or $5\frac{1}{2}$ inches. Swaging for 5 inches nominal diameter tube is $4\frac{1}{8}$ inches outside diameter. The length of the swaging is always 5 inches irrespective of the thickness of the tube sheet to which it is applied. The details of superheater tube swaging are shown by Fig. 3. After the swaging operation, the end of the tube should have all scale removed by light filing or grinding. In setting the superheater tube at the firebox end it should be so gaged as to allow $\frac{1}{4}$ inch for beading beyond the edge of the tube sheet. The general process of setting is similar to that used for the body tubes; the tube should be fitted neatly inside the ferrule, after the ferrule has been first tightened in the hole in the tube sheet.

Usually no preparation is required for the tube at the smokebox end other than cutting to correct length and the removal of scale by light filing or grinding. In special cases the tube may be expanded slightly at its front end, this expansion usually being limited to $\frac{1}{8}$ inch increase above the nominal diameter. This is a matter of convenience in removing the body of the tube through the hole and not necessarily a feature of correct tube application. When finally set, the tube should be of sufficient length to project from $\frac{1}{4}$ inch to $\frac{1}{2}$ inch beyond the face of the smokebox tube sheet. Beading of all tubes at the smokebox end is not necessary. The practice of The Baldwin Locomotive Works is to bead from 10 percent to 15 percent of these for purpose of better support to the tube sheet.

Diagrams and descriptions are submitted showing the various operations required for the correct setting of locomotive boiler tubes. The description of these operations assumes that the holes in the tube sheet have already been properly prepared in accordance with instructions previously given.

In the first operation, ferrules must be placed in the holes in the firebox tube sheet as shown by Fig. 4, then tightened with a roller expander or lightly set with a sectional expander. The edge of the copper ferrule should be set 1/32 inch inside the face of the sheet.

In the second operation, the tube end should be placed in the hole in the firebox sheet by fitting it neatly into the already placed copper ferrule. The usual beading allowance is $\frac{1}{4}$ inch, therefore the end of the tubes should project this amount beyond the surface of the tube sheet. It is suggested that the workmen employ a gage for this purpose so that uniformity can be obtained for the purpose of correct beading. After carefully setting into the ferrule, the tubes should be tightened with a roller expander or with a sectional expander as shown in Fig. 5.

Pneumatic operation of the roller expander will be found most convenient. An ordinary pneumatic drill of sufficient capacity can be used for this purpose, although a special driving tool can be procured if desired. The roller expander can also be operated by hand, the mandrel being turned carefully while maintaining even pressure on the expanding taper. Care should be used to not overwork the tube. Too much rolling is as bad as too little rolling. Operation of the sectional expander should be by a short-stroke pneumatic hammer or by a hand hammer. It is necessary to turn the expander in the tube between each consecutive driving in of the mandrel. One-quarter to one-half turn of the expander is recommended. Three or four driving operations are usually all that are required.

In the third operation, the front end of the tube should now be set. This operation is usually accomplished by the use of a roller expander as shown by Fig. 4. The projection at the front end is not so important as at the firebox tube sheet and exact gaging is not necessary, but the workmen should adhere to the limitations previously given.

In the fourth operation, all tubes (at both front and back ends) that are to be beaded should first be flared with a standard flaring tool. The tool recommended by The Baldwin Locomotive Works is shown by Fig. 6. This tool should be carefully used with a hand hammer or a short-stroke pneumatic hammer.

After completing the fourth operation, all tubes should be examined previous to beading, so that ends cracked in the expanding or flaring process may be detected. All tubes showing defects should be removed and replaced by sound ones.

The fifth operation of beading all tubes at their firebox end and the beading of selected tubes at the smokebox end should be carefully accomplished by means of a specially formed beading tool in the manner shown by Fig. 7. This should be done with a short-stroke pneumatic hammer or with a hand hammer. Care should be taken to see that nothing enters between the

the bolts with a hammer in order to determine whether they are broken. This test may be made either when the boiler is empty or when it has two or more gages of water under a pressure of at least 50 pounds, except that the testing of flexible bolts and crown bolts, immediately after the hydrostatic test, must be made while the boiler is under a hydrostatic pressure not less than the working pressure.

In addition to the hammer test, an inspection of all the visible tell-tale holes of the drilled bolts must be made to insure that they are open, using a drill for this purpose, if necessary. The firebox sheets must also be examined for indications of broken bolts, mud burns or bulging.

No boiler shall be permitted to remain in service when it has a broken crown bolt nor when there are one or more staybolts broken in the top row of the side, throat or fire door sheets, nor two or more adjacent broken or plugged staybolts in any part of the firebox, nor three or more broken or plugged staybolts in a circle of 4 feet in diameter, nor five or more broken in the entire boiler. Any broken or defective staybolt must be renewed as promptly as possible after discovery. Bolts with tell-tale holes leaking or plugged must be considered broken and renewed accordingly. Tell-tale holes must not be plugged except in cases of emergency. If a leak develops after an engine has been dispatched it may be plugged, if necessary, but the staybolt must be renewed upon arrival at the terminal.

The tell-tale holes of all staybolts must be kept open at all times, and must be thoroughly cleaned out with a 3/16-inch drill when locomotives are in shop for special examination of staybolts, and those visible must be cleaned out with a 3/16-inch drill prior to hydrostatic test.

Precaution must be taken to insure that all defective staybolts and crown bolts are removed. A careful examination must be made of all rigid and flexible staybolts, and crown bolts adjacent to those having defects, by cutting them out in all directions.

The caps of all expansion crown bolts in the two or more transverse rows at the front end of the roof sheet must be removed and the nuts adjusted each time a new set of flues is applied or the complete set of old flues are reset. The caps on these transverse rows of bolts must also be removed and the nuts adjusted at the first two monthly inspection periods following the application of a new firebox or the placing of a new boiler in service.

All locomotives must be given special staybolt examination, including the removal of caps from all flexible staybolts, and expansion crown bolts as required by Federal regulations, at each alternate class repairs, except that in every case when a locomotive is in shop for class repairs and the firebox or shell receives new half, three-quarters or full side sheets, it must be given special staybolt examination at that time and at the following alternate class repairs.

Locomotives that do not receive special staybolt examination within a period of two years must be removed from service and have the caps removed from all flexible staybolts and expansion crown stays as required by Federal regulation and as stated in the paragraph on flexible staybolt requirements below.

The two year period may be extended by the amount of time the locomotive has been out of service (full calendar months only to be counted), provided the total time between the removal of caps does not exceed three years. The master mechanic must arrange, if possible, to have the caps removed at the class repairs occurring in between the special staybolt examinations for those locomotives that will not, within the 24 months' period, make their mileage allotted to the alternate class repairs.

All hollow staybolts, that are behind grate bearers or stoker details and cannot be hammer tested from the inside of the firebox without removing grate bearers or stoker details, must be cut out. On locomotives equipped with duplex stokers, renew all staybolts in the back head that cannot be removed in the engine house without removal of the stoker elevators.

Hammer test all the remaining rigid staybolts in the usual manner as noted previously, and remove the defective bolts as specified.

Cut out 10 percent of all the remaining rigid bolts (not including crown bolts), selecting the majority from the zones that are known to have frequent staybolt failures, particularly those in the lower portion of the back head, the upper corners of the back head, the four corners of the side sheet, the lower corners of the throat sheet and those in other locations which in the judgment of the inspector should be given special consideration. In radial stay boilers, all staybolts that are 14 inches or more in length should be regarded as crown bolts.

Before making the special examination of rigid staybolts, the boiler inspector must obtain from the point where the permanent record is kept, a copy of the diagrams showing the results of the last special staybolt examination and all the rigid and flexible staybolts that were removed since that examination. This copy of the diagrams must be prepared from the permanent file record of the superintendent of motive power by the office which is designated by him to have charge of the same and must be forwarded promptly to the shop which is to make the special examination. The boiler inspector must determine from the diagram the relative ages of the staybolts and the localities that have had extensive staybolt failures. The rigid staybolts that were not cut out at or since the last special examination and also the rigid staybolts in the localities having had extensive failures, must be given special attention. In every case a competent boiler inspector must select the staybolts to be removed.

In cutting out the rigid bolts that are selected for removal in this special examination, the staybolts must be cut loose from the inside firebox sheet and then bent away from the inside sheet to determine the degree of fracture, if any, before cutting them loose from the outside sheet.

All flexible staybolts (including expansion crown stays) must have their caps removed and be sounded with a hammer, first from the outside and then from the inside end, and the defective bolts removed, and also the adjacent bolts in each direction until sound bolts are found. The expansion crown bolts and nuts must be examined and the loose nuts, or the bolts too short to have full thread in the nut, must be renewed. Bolts from which nuts have been removed should be checked with proper gages before new nuts are applied.

Cut out 10 percent of the flexible staybolts in the top corners of the throat sheet.

Cut out of the side sheets 10 percent of the flexible staybolts located in the top longitudinal row and 10 percent of the flexible staybolts in the outside vertical row at the front and back from the top down to the center line of boiler.

Ten percent of the flexible staybolts located in the 14 bottom longitudinal rows of the cylindrical combustion chamber from the flue sheet back to and including the row of bolts below the arch tube holes must be cut out and examined at both ends to determine the condition of the bolts in that locality.

# Questions and Answers Pertaining to Boilers

**This department is maintained for the purpose of helping those who desire assistance on practical boiler shop problems. Inquiries should bear the name and address of the writer. Anonymous communications will not be considered. The identity of the writer, however, will not be disclosed unless the editor is given special permission to do so.**

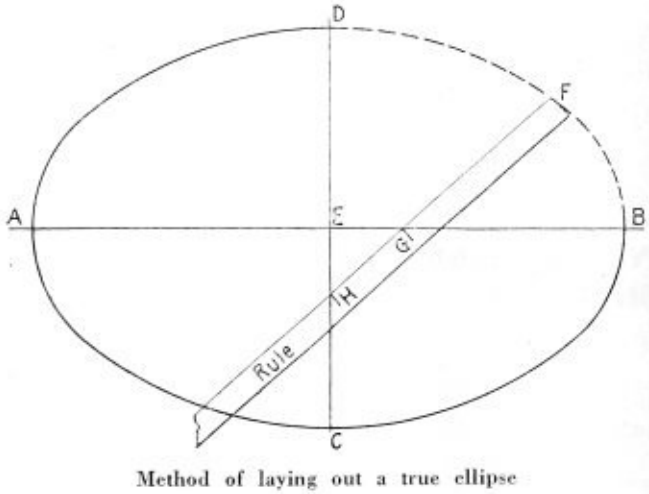
## By George M. Davies

### Layout of Ellipse

Q.—Will you tell me through the Questions and Answers Department of THE BOILER MAKER an accurate and simple way to draw an ellipse, such as a 10-inch by 15-inch manhead? H. H.

A.—An ellipse is a conic section. No part of an ellipse is a part of a circle, therefore it cannot be drawn with the ordinary compasses. A quick and accurate method of drawing an ellipse is as follows:

Draw the lines *A-B* and *C-D* at right angles to each other. Set off from *E* to *B* and *E* to *A* half the major




**Method of laying out a true ellipse**

axis, or 7½ inches; set off from *E* to *C* and *E* to *D* half the minor axis, or 5 inches. Now mark on your rule the length *F-G* equal to *E-D* and mark *F-H* equal to *E-B*, then by moving the rule around and always keeping the point *H* on the line *C-D*, and the point *G* always on the line *A-B*, the point *F* will be exactly upon the curve of a true ellipse.

### Removal of Staybolt Caps

Q.—What are the reasons for removing the caps from flexible staybolts on a locomotive? I would like to have all the reasons you know of.—J. M. D.

A.—The reasons for removing the caps from flexible staybolts on locomotive boilers are as follows:

(1) To renew a broken staybolt.
(2) To renew a fractured sleeve.
(3) To inspect the staybolt.
(4) To remove and reapply staybolts when applying new firebox sheets.
(5) To replace a defective cap or cap gasket.
(6) To comply with the following I.C.C. Rules.

Rule 23. Method of testing flexible staybolts with caps.

Except as provided in paragraph (b) all staybolts having caps over the outer ends shall have the caps removed at least once every two (2) years and the bolts and sleeves examined for breakage. Each time the hydrostatic test is applied the hammer test required by rules 21 and 22 shall be made while the boiler is under hydrostatic pressure not less than the allowed working pressure.

(b) When all flexible staybolts with which any boiler is equipped are provided with telltale holes not less than 3/16 inch nor more than 7/32 inch in diameter, extending the entire length of the bolt and into the head not less than one-third of its diameter, and these holes are protected from becoming closed by rust and corrosion by copper plating or other approved method, and are opened and tested, each time the hydrostatic test is applied, with an electrical or other instrument approved by the Bureau of Locomotive Inspection, that will positively indicate when the telltale holes are open their entire length, the caps will not be required to be removed. When this test is completed the hydrostatic test must be applied and all staybolts removed which show leakage through the telltale hole.

The inner ends of the telltale holes must be kept closed with fireproof porous material that will exclude foreign matter and permit leakage of steam or water, if the bolt is broken or fractured, into the telltale hole.

When this test is completed the ends of the telltale holes shall be closed with material of different color than that removed and a record kept of colors used.

(c) The removal of flexible staybolt caps and other tests shall be reported on the report of inspection Form No. 3 and a proper record kept in the office of the railroad company of the inspections and tests made.

(d) Firebox sheets must be carefully examined at least once every month for mud, burn, bulging and indications of broken staybolts.

(e) Staybolt caps shall be removed or any of the above tests made whenever the United States inspector considers it desirable in order to thoroughly determine the condition of staybolts or staybolt sleeves.

Requirements of Rule 23 may be interpreted as follows:

(a) When locomotives are stored for one or more full calendar months, the removel of flexible staybolt caps for the purpose of inspection will be due after 24 calendar months service, provided such service is performed within three consecutive years. Portions of calendar months out of service will not be counted. Time out of service must be properly accounted for by out-of-service reports filed with the United States inspector and notation of months claimed out of service must be made on the back of each subsequent inspection report and cab card.

(b) A locomotive stored when flexible staybolt cap removals are due need not be given such test until immediately prior to being returned to service.

(c) When locomotives are being given their annual inspection and test, flexible staybolt caps should be removed at the time of making this inspection, if they become due for removal before another annual inspection is due, and their removal shown on Form 3. One of the principal reasons urged for extending the time for removal of flexible staybolt caps from 18 months to two years was for the purpose of bringing the test due at every second annual inspection.